

Template Screening Report (v5.7.1 - Multi-Source AI Evaluator)

Journal Template: Jurnal Teknologi

Submission No: 26813

Author Name: SAlI

Article Title: Comparative Analysis of Laplace and Sumudu Transform Techniques in Modeling and Simulation of DC Motor Transfer Function

Source File: Intisar.docx

Screened PDF: Intisar.pdf

Screening Date: 2026-05-21 09:43:20

Detection Source Debug Report

Item	DOC Detection	PDF Detection	AI Detection	Final Source
Article Title	NO	YES	SKIPPED	PDF only
Authors	NO	YES	SKIPPED	PDF only
Affiliations	NO	YES	SKIPPED	PDF only
Corresponding Author	NO	NO	SKIPPED	None
Graphical Abstract	NO	NO	SKIPPED	None
English Abstract	NO	YES	SKIPPED	PDF only
Keywords	NO	NO	SKIPPED	None
Malay Abstract (Abstrak)	NO	YES	SKIPPED	PDF only
Kata Kunci	NO	NO	SKIPPED	None
Introduction	YES	YES	SKIPPED	DOC + PDF
Methodology	NO	NO	SKIPPED	None
Results and Discussion	YES	YES	SKIPPED	DOC + PDF
Conclusion	YES	NO	SKIPPED	DOC only
Acknowledgement	NO	NO	SKIPPED	None
References	YES	NO	SKIPPED	DOC only

Summary: DOC detected 4 items | PDF detected 7 items | AI confirmed 0 items

The system uses DOC + PDF + AI voting logic. AI serves as a third checker when DOC and PDF conflict. AI is skipped when DOC and PDF already agree.

Executive Summary

Overall Decision: **NEEDS MANUAL REVIEW**

Main Reason: The manuscript generally follows the template but contains inconsistencies that require manual review.

Critical Failed Items:

- Graphical Abstract
- Methodology

Items Needing Review:

- Article Title
- Authors
- Affiliations
- English Abstract
- Malay Abstract (Abstrak)

Suggested Action: Editor should review the highlighted issues before proceeding.

Overall Result

Metric	Score / Value
Final Status	NEEDS MANUAL REVIEW
Final Score	53/100
Structure Score	20%
Layout Score	56%
Formatting Score	93%
Detection Method	DOCX-First + Multi-Source AI (v5.7.1)
Decision	MANUAL REVIEW

How Items Are Evaluated

DOC + PDF + AI SKIPPED	PASS	Both deterministic detectors agree
DOC + PDF + AI CONFIRMED	PASS	All three confirm
DOC + AI	NEEDS REVIEW	DOC+AI confirm, but PDF layout conflicts
PDF + AI	NEEDS REVIEW	PDF+AI confirm, but DOC source extraction fails
DOC only	NEEDS REVIEW	DOC detects, but PDF+AI don't confirm
PDF only	NEEDS REVIEW	PDF detects, but DOC+AI don't confirm
AI only	NEEDS REVIEW	Only AI confirms (deterministic detectors failed)
None	FAILED	All three fail to detect required item

Status Explanation

PASS — Item follows the template structure, position, and formatting.

WARNING/NEEDS REVIEW — Item exists but evidence is inconsistent or layout differs from template.

FAILED — Required item truly missing after DOC/PDF/AI checks.

Item Screening Details

Item	DOC	PDF	AI	Final Source	Status	Reason
Article Title	NO	YES	SKIPPED	PDF only	WARNING	WARNING detected in PDF layout, but DOC and AI did not confirm
Authors	NO	YES	SKIPPED	PDF only	WARNING	WARNING detected in PDF layout, but DOC and AI did not confi

Affiliations	NO	YES	SKIPPED	PDF only	WARNING	Affiliations detected in PDF layout, but DOC and AI did not confirm.
Corresponding Author	NO	NO	SKIPPED	None	FAILED	Corresponding_Author not found by DOC, PDF, or AI.
Graphical Abstract	NO	NO	SKIPPED	None	FAILED	Graphical_Abstract not found by DOC, PDF, or AI.
English Abstract	NO	YES	SKIPPED	PDF only	WARNING	Abstract detected in PDF layout, but DOC and AI did not confirm.
Keywords	NO	NO	SKIPPED	None	FAILED	Keywords not found by DOC, PDF, or AI.
Malay Abstract (Abstrak)	NO	YES	SKIPPED	PDF only	WARNING	Abstract detected in PDF layout, but DOC and AI did not confirm.
Kata Kunci	NO	NO	SKIPPED	None	FAILED	Kata_Kunci not found by DOC, PDF, or AI.
Introduction	YES	YES	SKIPPED	DOC + PDF	PASS	Introduction confirmed by both DOC and PDF detection.
Methodology	NO	NO	SKIPPED	None	FAILED	Methodology not found by DOC, PDF, or AI.
Results and Discussion	YES	YES	SKIPPED	DOC + PDF	PASS	Results_Discussion confirmed by both DOC and PDF detection.
Conclusion	YES	NO	SKIPPED	DOC only	WARNING	Conclusion detected in DOC, but PDF and AI did not confirm.
Acknowledgement	NO	NO	SKIPPED	None	PASS	Acknowledgement not found by DOC, PDF, or AI.
References	YES	NO	SKIPPED	DOC only	WARNING	References detected in DOC, but PDF and AI did not confirm.

Detailed Findings

Article Title [title]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Article Title' section must exist.
Found: Matched heading: "Comparative Analysis of Laplace and Sumudu Transform Techniques in Modeling and Simulation of DC"
| Detection method: score_ranking
Confidence: 0.95
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: PDF
Final Source Used: PDF only
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, top area
Found: Page 1, left column, top area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Heading should be in uppercase style.
Expected: Heading should use bold font.
Formatting Check: WARNING
Reason: Formatting differs from template expectations.
Overall Item Status: WARNING

Authors [authors]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Authors' section must exist.
Found: Matched heading: "Among the most important advancement that characterize multipurpose tools for solving differential e" |
Detection method: score_rank
Confidence: 0.70
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: PDF
Final Source Used: PDF only
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Affiliations [affiliations]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Affiliations' section must exist.

Found: Matched heading: "Among the most important advancement that characterize multipurpose tools for solving differential e" |

Detection method: score_rank

Confidence: 0.50

Detection Method: score_ranking

DOC Detection: NO

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: PDF

Final Source Used: PDF only

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Page 1, left column, upper area

Position Check: PASS

Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Corresponding Author [corresponding_author]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Corresponding Author' section must exist.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO

AI Detection: SKIPPED

Evidence Sources: None

Final Source Used: None

Status: FAILED

Reason: Required component 'Corresponding Author' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Graphical Abstract [graphical_abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Graphical Abstract' component must exist.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO

AI Detection: SKIPPED

AI Evidence: AI rate limit reached (max calls exceeded)

Evidence Sources: None

Final Source Used: None

Status: **FAILED**

Reason: Required component 'Graphical Abstract' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, upper area

Found: Not detected

C. Formatting Evaluation

Expected: Image should be present.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **FAILED**

English Abstract [abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'English Abstract' section must exist.

Found: Nearby content: "To address this, a semi-analytical approach combining the Elzaki integral transform and the projected differential transform method

Confidence: 0.00

Detection Method: semantic_fallback

DOC Detection: NO

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: PDF

Final Source Used: PDF only

Status: **WARNING**

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: **PASS**

Reason: Formatting matches template expectations.

Overall Item Status: **WARNING**

Keywords [keywords]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Keywords' section must exist.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO

AI Detection: SKIPPED

Evidence Sources: None

Final Source Used: None

Status: **FAILED**

Reason: Required component 'Keywords' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, lower-middle area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Malay Abstract (Abstrak) [abstrak]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Malay Abstract (Abstrak)' section must exist.

Found: Nearby content: "To address this, a semi-analytical approach combining the Elzaki integral transform and the projected differential transform method

Confidence: 0.00

Detection Method: semantic_fallback

DOC Detection: NO

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: PDF

Final Source Used: PDF only

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Kata Kunci [kata_kunci]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Kata Kunci' section must exist.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO

AI Detection: SKIPPED

Evidence Sources: None

Final Source Used: None

Status: FAILED

Reason: Required component 'Kata Kunci' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Introduction [introduction]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Introduction' section must exist.
Found: Matched heading: "Introduction" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: **PASS**
Reason: The required component 'Introduction' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, left column, upper-middle area
Position Check: **WARNING**
Reason: Position deviates from expected region (tolerance: 0.15).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: **PASS**
Reason: Formatting matches template expectations.
Overall Item Status: **PASS**

Methodology [methodology]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Methodology' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: None
Final Source Used: None
Status: **FAILED**
Reason: Required component 'Methodology' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: **PASS**
Reason: Formatting matches template expectations.
Overall Item Status: **FAILED**

Results and Discussion [results_discussion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Results and Discussion' section must exist.
Found: Matched heading: "Result and Discussion" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: PASS

Reason: The required component 'Results and Discussion' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Page 1, left column, upper-middle area

Position Check: WARNING

Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: PASS

Conclusion [conclusion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Conclusion' section must exist.

Found: Matched heading: "Conclusion" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: NO

AI Detection: SKIPPED

Evidence Sources: DOC

Final Source Used: DOC only

Rejected false positives: 7 Conclusion

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Acknowledgement [acknowledgement]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Acknowledgement' component is recommended.

Found: Detection method: not_found

Confidence: 0.00

Detection Method: not_found

DOC Detection: NO

PDF Detection: NO

AI Detection: SKIPPED

Evidence Sources: None

Final Source Used: None

Status: PASS

Reason: The required component 'Acknowledgement' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

References [references]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'References' section must exist.
Found: Matched heading: "References" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: DOC
Final Source Used: DOC only
Rejected false positives: 8 References
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Editor Conclusion

The manuscript generally follows the template but has items that require manual review. These items had inconsistent evidence between DOC detection, PDF layout analysis, or AI verification. Please review the highlighted items before making a final decision.

Structure Score: 20% (Weight: 40%)

Layout Score: 56% (Weight: 30%)

Formatting Score: 93% (Weight: 30%)

Final Score: 53/100

Critical Items Failed:

- Graphical Abstract — **FAILED**
- Methodology — **FAILED**

Items Needing Review:

- Article Title — **WARNING**
- Authors — **WARNING**
- Affiliations — **WARNING**
- English Abstract — **WARNING**
- Malay Abstract (Abstrak) — **WARNING**

This report was automatically generated by Turnitin Assistant v5.7.1 Template Screening Engine (DOCX-First + Multi-Source AI Evaluator).

Comparative Analysis of Laplace and Sumudu Transform Techniques in Modeling and Simulation of DC Motor Transfer Function

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BSTRACT

Among the most important advancement that characterize multipurpose tools for solving differential equations and analyzing complex systems are Laplace Transform (LT) and Sumudu Transform (ST). Adapting differential equations into algebraic forms to simplify the solution of ordinary linear differential equations and analysis the linear dynamic systems is the aim of this work. LT and ST are applied as a powerful investigative tool. The fundamental principles and properties of these two transformations presented in this study, and explained their interrelations depends on their comparative analysis between analytical solutions for dynamic behavior of a DC motor by using the LT and ST with numerical solutions using MATLAB. From the electrical and mechanical equations of the motor, the mathematical model derived. The relationship between input voltage and angular speed represented by obtained the transfer function of it. The potential of ST as a substitute analytical tool to LT in control system analysis as the conclusion after the accuracy is evaluated by using RMSE and percentage error. In spite of the LT domain remains more physically intuitive for the control system, the ST provides an analytically simpler or mathematically smoother framework for solving differential equations.

1 Introduction

Solving differential equations required intricate mathematical manipulations in traditional methods, and some of it not applicable to nonlinear or time-varying systems. Methods of different transformations offer solutions to these problems by providing a systematic framework to convert differential equations to simpler formulae which can be easily solved by using algebraic techniques. The powerful mathematical tools that have revolutionized the fields of mathematical physics and engineering systems are: Laplace Transform (LT) and Sumudu Transform (ST). Every one of these two transformations offers unique competences to solve the differential equations, analyze the dynamic systems, and model complex phenomena. In this section, a brief overview for LT and ST presented with discuss the motivations behind their development, and focused their significance in mathematical physics and engineering. To formulate the mathematical models which accurately describe the behavior of physical systems, and predict their development over time the researchers investigated these transformations. Usually, LT uses integral transform which converts functions of time to functions of complex frequency to simplify the process of solving complex dynamical systems [1]. Recently, ST is a relatively addition to the family of integral transforms that operates in the time-domain while LT operates in the complex frequency domain. ST is useful to analyze time-varying signals and systems [2]. It has several advantageous properties, like including translation, linearity, and convolution properties which make it a valued tool to solve differential and integral equations that become apparent in mathematical physics and engineering [3]. The comprehensive introduction to Laplace-Sumudu-Elzaki Transformation (LSET) provided in previous research, and different applications for the realm of mathematical physics explored [4]. Khan et al. presented the modeling physical, analyzed the dynamical behavior, and predicted system responses by using LT. Unsteady agnetohydrodynamic flow for the second grade tetra hybrid nanofluid in a porous medium containing uniformly discrete dust units predicted. Furthermore, effectiveness of LT and STs in analyzing and simulating the turbulent MHD flow of nanofluids through absorbent media with dispersed dust atoms demonstrated. In addition, a novel methodology developed for solving nonlinear fractional partial differential equations. It combined the conformable double Laplace ST (DLST) with the adapted decomposition technique to form the conformable double LT-ST modified decomposition (CDLSMD) method [5]. Another study focused on the efficiency of the double Laplace-ST (DLST) to solve the partial differential equations. They proved the theorems regarding the properties of DLST, including the convolution theorem that assists the solution of partial differential equations [6]. The population balance equations that prevalent in different fields like chemical study and planetary science posed important challenges due to model complexity.

To address this, a semi-analytical approach combining the Elzaki integral transform and the projected differential transform method has been proposed. This method offers direct solutions to linear or non-linear differential equations without discretization or linearization, enhancing accuracy and convergence in solving aggregation, aggregation breakage, and multi-dimensional aggregation equations [4]. On the other hand, important results of conformable fractional partial derivatives presented, and the conformable triple Laplace and ST were coupled with the Adomian decomposition method where a new method proposed to solve nonlinear partial differential equations in 3-space [3]. Furthermore, the variational iteration method for computing of the LT and ST of functions applied by using a different way. Unlike the literature and also classical computation the results clearly showed that method provided a powerful and easy calculation, and not required too long operations [7].

In engineering, the Laplace and STations play a crucial role in designing control systems, analyzing electronic circuits, optimizing signal processing algorithms, and simulating complex phenomena for improving the performance and enhancing the reliability of the engineering systems in different applications [8], [9].

The properties, applications, and computational techniques that associated with these transformations investigated in the subsequent sections of this work to provide the insights into their applied utility and theoretical basics.

2 Mathematical Modeling of DC Motor

One of the common actuator used in control systems is the DC motor. It directly provides rotary motion and can produce translational motion when coupled with wheels or drums and cables. DC motors are preferred in control systems because they provide continuous motion unlike the stepper motor, which produces motion in steps. A mathematical model of a DC motor is derived to analytically observe and define the control mechanism [10]. Accurate model building is a crucial step in practical control problems. An adequately developed system model is essential for the reliability of the designed control. The system parameters should be determined utilizing system identification techniques. However, an appropriate model structure should be found before the identification step can be implemented [7].

To model the DC servo motors, governing differential equations, which express the motor characteristics and relate the applied voltage to the torque produced by the rotor, are needed. The manufacturers provide some of the initial physical parameters, such as the motor torque T , armature current i and the constant factor K_t , as shown as follows [11]:

$$T = K_t \cdot i$$

Mathematical

Modeling of

DC Motor..

eq.1

(e) is the back-EMF that proportional to the angular velocity of the shaft (ω) by a constant factor K_e :

$$e = K_e \cdot \omega$$

Mathematica

l Modeling of

DC Motor..

eq.2

In SI units, the motor torque and back-EMF constants are equal, that is,

$$k = K_t = K_e$$

The system is modelled by summing the torques that act on the rotor inertia and integrate the angular acceleration of the rotor to provide the velocity and integrate the velocity to obtain the position. Kirchhoff's laws are also applied to the armature circuit. First, a modelling of the integrals of the rotor acceleration and of the rate of change of armature current is calculated, as shown follows [12][13]:

$$\iint \frac{d^2\theta}{dt^2} dt = \int \frac{d\theta}{dt} dt = \theta$$

Mathematica

l Modeling of

DC Motor..

eq.3

Mathematica

l Modeling of

DC Motor..

eq.4

Next, from the mechanical part of DC motor circuit, Newton's and Kirchhoff's laws are applied to the motor system, and substitute eq.3 in it to generate the following equations:

$$J \frac{d^2\theta}{dt^2} + b \frac{d\theta}{dt} = T \Rightarrow \frac{d^2\theta}{dt^2} = \frac{1}{J} (k \cdot i - b \frac{d\theta}{dt})$$

Mathematical

Modeling of

DC Motor..

eq.5

Then, the equation for the electrical part of DC motor generated as:

$$L \frac{di}{dt} = -Ri + v - e \Rightarrow \frac{di}{dt} = \frac{1}{L} (-Ri + v - k \frac{d\theta}{dt})$$

Mathematical

Modeling of

DC Motor..

eq.6

where:

R = Armature resistance [Ohm]

L = Armature inductance [Henry]

k = Back emf constant [V/(rad/s)]

J = Rotor inertia [kg.m²]

B = Rotor damping [N.m/(rad/s)]

In mathematical analysis, LT and ST stand as formidable tools by presenting unique approaches to solve the differential equations and demonstrating complex systems. Essential the effective application in the mathematical physics and engineering is understand the theoretical basics and fundamental properties of these transformations.

3 Laplace Transform

LT named after the French mathematician Pierre-Simon Laplace, it is a powerful integral transform which change the time functions to complex frequency functions. It is signified by the operator $f(t) = F(s)$, where: $f(t)$ is the input function, $F(s)$ is the LT for the input function, and s is a complex variable. The LT uses to solve linear ordinary and partial differential equations by converting them into algebraic equations. Basic properties of the LT included linearity, frequency shifting, time shifting, convolution, integration, differentiation, and initial value theorem. So, a wide range of mathematical problems in applied mathematics, physics, and engineering is using the LT to handle it. In addition, it is suitable for analyzing dynamical systems, theory of control, electrical circuits, and signal processing applications. Laplace Variable is (s) which is used to convert the signals and differential equations from the time domain (t) to a composite domain named (composite frequency domain / complex frequency domain). It characterizes the response of a system to give the signal input, it is usually derive from differential equations and transform to easily solvable algebraic equations in (s) domain by using the LT.

Laplace

$$s = j\omega + \sigma$$

Transform..

eq.7

where:-

j = imaginary unit ($j = \sqrt{-1}$)

ω = the imaginary part, representing the angular frequency of the signal

σ = the real part, representing damping or exponential growth

Laplace Analysis of DC Motor

The transfer function is a mathematical statement that expresses the transfer characteristics of a system or equipment, and it is named $G(s)$. It is the relationship between the rotational speed (output of the system) and the armature voltage (output of the system), as shown as follows:

Laplace

$$G(s) = \frac{Out(s)}{In(s)} = \frac{\omega(s)}{V(s)}$$

Transform..

eq.8

$$J \frac{d^2 \theta}{dt^2} + b \frac{d\theta}{dt} = k \cdot i$$

$$\mathcal{L} \left\{ J \frac{d^2 \theta}{dt^2} + b \frac{d\theta}{dt} \right\} = \mathcal{L} \{ ki \}$$

$$J \mathcal{L} \left\{ \frac{d^2 \theta}{dt^2} \right\} + b \mathcal{L} \left\{ \frac{d\theta}{dt} \right\} = \mathcal{L} \{ ki \}$$

$$J \mathcal{L} \left\{ \frac{d^2 \theta}{dt^2} \right\} + b \mathcal{L} \left\{ \frac{d\theta}{dt} \right\} = \mathcal{L} \{ ki \}$$

$$J \left[s^2 \Theta(s) - s\theta(0) - \theta'(0) \right] + b \left[s\Theta(s) - \theta(0) \right] = kI(s)$$

$$J s^2 \Theta(s) + b s \Theta(s) = kI(s)$$

$$s\Theta(s)(Js+b) = kI(s)$$

Laplace

Transform..

eq.9

$$L \frac{di}{dt} + Ri = v - k \frac{d\theta}{dt}$$

$$\mathcal{L} \left\{ L \frac{di}{dt} + Ri \right\} = \mathcal{L} \left\{ v - k \frac{d\theta}{dt} \right\}$$

$$\mathcal{L} \left\{ \frac{di}{dt} \right\} + R \mathcal{L} \{ i \} = \mathcal{L} \{ v \} - k \mathcal{L} \left\{ \frac{d\theta}{dt} \right\}$$

$$\mathcal{L} \{ sI(s) - i(0) \} + RI(s) = V(s) - k \{ s\Theta(s) - \theta(0) \}$$

$$\mathcal{L} s I(s) + R I(s) = V(s) - k s \Theta(s)$$

$$(Ls + R) I(s) = V(s) - k s \Theta(s)$$

Laplace

Transform..

eq.10

$$I(s) = \frac{V(s) - k s \Theta(s)}{Ls + R}$$

Substitute eq.9 in eq.10 as follow:

$$s\Theta(s)(Js+b) = k \left(\frac{V(s) - k s \Theta(s)}{Ls + R} \right) = \frac{kV(s) - k^2 s \Theta(s)}{Ls + R}$$

$$s\Theta(s)(Js+b)(Ls+R) = kV(s) - k^2 s \Theta(s)$$

$$kV(s) = s\Theta(s)(Js+b)(Ls+R) + k^2 s \Theta(s)$$

$$kV(s) = \Theta(s) \{ s(Js+b)(Ls+R) + k^2 s \}$$

$$\frac{\Theta(s)}{V(s)} = \frac{k}{s(Js+b)(Ls+R) + k^2 s}$$

$$G(s) = \frac{\Theta(s)}{V(s)} = \frac{k}{s[(Js+b)(Ls+R)+k^2]}$$

Laplace
Transform..
eq.11

4 Sumudu Transform

Introduced by Professor K. M. Saad Soliman in 1994, the ST is a relatively new integral transform that operates in the time-domain, unlike the LT which operates in the complex frequency domain. The ST of a function $f(t)$ is denoted by $f(t) = F(u)$, where u is a real variable. Sumudu method has many possesses features which is suitable using in systems analysis. The most important characteristics of that are:

1. Preserves the function's units.
2. Suitable for solving linear / nonlinear differential equations.
3. Directly relates the time domain to the time modulo without complex transformations like Laplace.

Sumudu Analysis of DC Motor

$$J \frac{d^2 \theta}{dt^2} + b \frac{d\theta}{dt} = k \cdot i$$

$$S \left\{ J \frac{d^2 \theta}{dt^2} + b \frac{d\theta}{dt} \right\} = S \{ ki \}$$

$$J S \left\{ \frac{d^2 \theta}{dt^2} \right\} + b L S \left\{ \frac{d\theta}{dt} \right\} = S ki \} \textcolor{red}{\downarrow}$$

$$J S \left\{ \frac{d^2 \theta}{dt^2} \right\} + b S \left\{ \frac{d\theta}{dt} \right\} = S ki \} \textcolor{red}{\downarrow}$$

$$J \left\{ \frac{1}{u^2} \Theta(u) - \frac{1}{u^2} \theta(0) - \frac{1}{u} \theta'(0) \right\} + b \left\{ \frac{1}{u} \Theta(u) - \frac{1}{u} \theta(0) \right\} = kI(u)$$

$$J \frac{1}{u^2} \Theta(u) + b \frac{1}{u} \Theta(u) = kI(u)$$

$$\frac{1}{u} \Theta(u) \left(J \frac{1}{u} + b \right) = kI(u)$$

Sumudu
Transform..
eq.12

$$L \frac{di}{dt} + Ri = v - k \frac{d\theta}{dt}$$

$$S \left\{ L \frac{di}{dt} + Ri \right\} = S \left\{ v - k \frac{d\theta}{dt} \right\}$$

$$L S \left\{ \frac{di}{dt} \right\} + R S \{ i \} = S \{ v \} - k S \left\{ \frac{d\theta}{dt} \right\}$$

$$L\left\{\frac{1}{u}I(u)-\frac{1}{u}i(0)\right\}+RI(u)=V(u)-k\left\{\frac{1}{u}\Theta(u)-\frac{1}{u}\theta(0)\right\}$$

$$L\frac{1}{u}I(u)+RI(u)=V(u)-k\frac{1}{u}\Theta(u)$$

$$\left(L\frac{1}{u}+R\right)I(u)=V(u)-k\frac{1}{u}\Theta(u)$$

$$I(u)=\frac{V(u)-k\frac{1}{u}\Theta(u)}{L\frac{1}{u}+R}$$

Sumudu

Transform..

eq.13

Substitute eq.12 in eq.13 to get transfer function as follow:

$$\frac{1}{u}\Theta(u)\left(J\frac{1}{u}+b\right)=k\left(\frac{V(u)-k\frac{1}{u}\Theta(u)}{L\frac{1}{u}+R}\right)=\frac{kV(u)-k^2\frac{1}{u}\Theta(u)}{L\frac{1}{u}+R}$$

$$\frac{1}{u}\Theta(u)\left(J\frac{1}{u}+b\right)\left(L\frac{1}{u}+R\right)=kV(u)-k^2\frac{1}{u}\Theta(u)$$

$$kV(u)=\frac{1}{u}\Theta(u)\left(J\frac{1}{u}+b\right)\left(L\frac{1}{u}+R\right)+k^2\frac{1}{u}\Theta(u)$$

$$kV(u)=\Theta(u)\left\{\frac{1}{u}\left(J\frac{1}{u}+b\right)\left(L\frac{1}{u}+R\right)+k^2\frac{1}{u}\right\}$$

So, transfer function is:

$$G(u)=\frac{\Theta(u)}{V(u)}=\frac{k}{\frac{1}{u}\left\{\left(J\frac{1}{u}+b\right)\left(L\frac{1}{u}+R\right)+k^2\right\}}$$

Sumudu

Transform..

eq.14

5 Numerical Comparison

The relation between the LT and ST lies in their ability to complement each other in different problem-solving scenarios. While the LT is well-suited for handling linear systems and boundary value problems, the ST offers an alternative approach for analyzing time-varying systems and non-linear phenomena.

In the LT, the symbol (s) is the transformation variable, which is the equivalent of the symbol (u) in the ST. But there is an important difference in meaning and characteristics:

1- Mathematical Definition, the ST is defined as follows:

$$S[f(t)]=F(u)=\int_0^{\infty} e^{-t} f(ut) dt$$

Where:

$f(t)$ = original function with time.

$F(u)$ = transformation function in the new domain.

u = the transformation variable in Sumodo (similar to (s) in Laplace).

- 2- The difference between (u) and (s) presents in table (1) as shown below:

Table (1) – comparison of transformation variable in Laplace and Sumodo

Comparison	Laplace Transform	Sumudu Transform
Formula	$L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$	$S[f(t)] = F(u) = \int_0^{\infty} e^{-t} f(ut) dt$
Variable	s for the inverse time (1/t)	s for the time (t)
Relationship	$s = \frac{1}{u}$	$u = \frac{1}{s}$
Output Domain	complex frequency domain	scaled time domain
Common Application	Control systems, Electrical circuits, dynamics	Analysis of systems, differential equations, special transforms

- 3- The useful relationship between them, the approximate relationship between the two transformations can be written as follows:

$$S[f(t)](u) = \left(\frac{1}{u}\right) L[f(t)] \frac{1}{u}$$

That is, the Sumodo transform is a modified form of the LT, but it keeps the same units of time in the resulting function.

6 Result and Discussion

In this study, the dynamic behavior of a DC motor was analyzed using two different integral transforms: the LT and the ST. Both transforms were applied to the same mathematical model derived from the electrical and mechanical equations of the DC motor (eq.11 for LT and eq.14 for ST) by using the parameters as shown in table (2) as follow:

Table (2) – Parameters DC motor

Parameter	Optimized value	unit
Armature Resistance (R)	4.0145	ohm
Armature Inductance (L)	3.2555 e ⁻⁴	H
Back EMF constant (K _m)	0.0216	V/rad/sec
moment of inertia of the rotor (J _d)	0.0183	kg.m ²
Rotor damping (b)	0.3786	Nms
Rotor inertia (J)	0.0183	kg.m ²

Firstly, the comparison between the step responses of the DC motor obtained using the analytical transfer function derived from the LT and the numerical simulation based on solving the differential equations directly as shown in Figure (1).

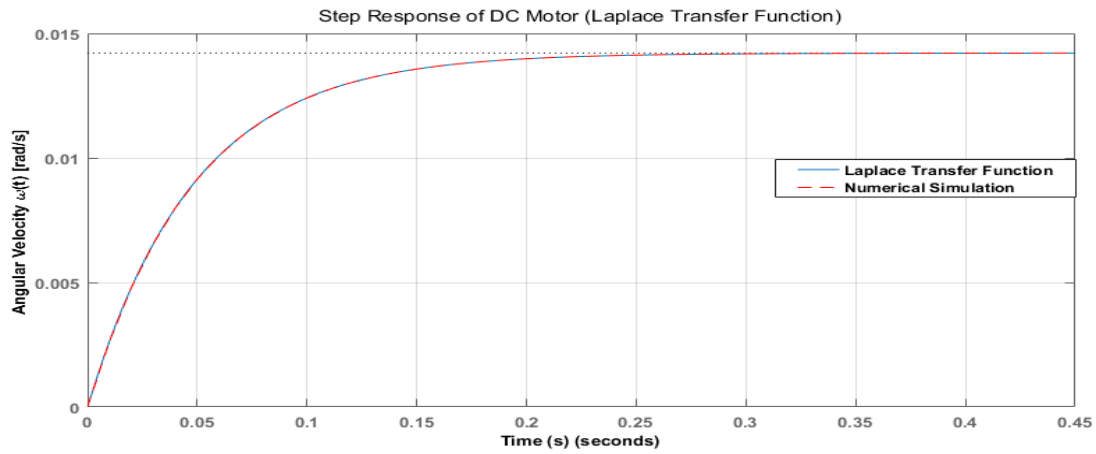


Figure (1) – Step response of DC motor for Laplace transfer function

In figure (1), the blue curve represented the analytical response which calculated from the transfer function eq.11, while the red dashed line corresponds to the numerical simulation results obtained using MATLAB's `ode45` solver [14]. It is observed that both responses almost overlap, indicating that the analytical model accurately represents the dynamic behavior of the motor. The close agreement between the two responses validates the correctness of the transfer function derivation and confirms the equivalence between the Laplace-domain and time-domain formulations of the system. This consistency also demonstrates that the system parameters (R , L , J , B , K_t , K_e) appropriately selected and that the Laplace-based analytical approach can reliably predict the transient and steady-state performance of the DC motor without the need for full numerical simulation [13], [15]. Then, analytical transfer function derived from the ST compared with the same numerical simulation which mathematical derivation was used for equivalent transformation function previous used with Laplace as shown in figure (2);

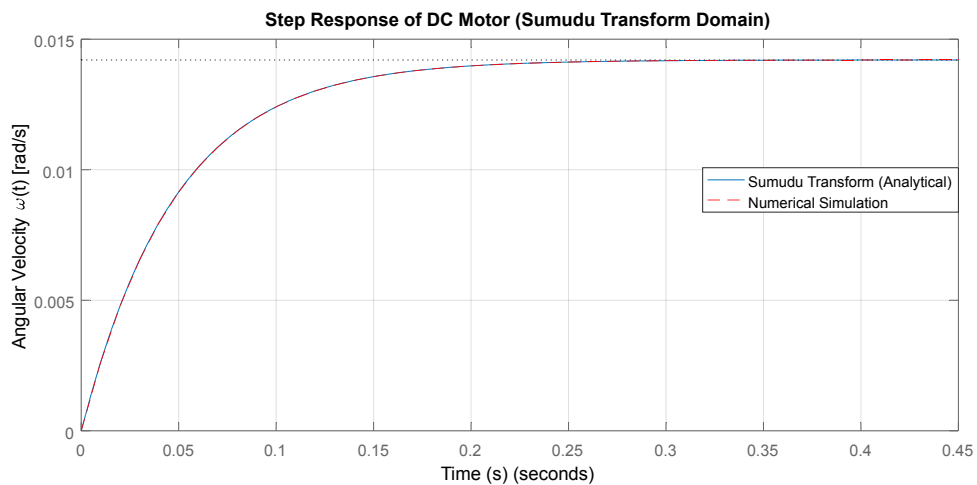


Figure (2) – Step response of DC motor for Sumudu transfer function

In MATLAB software, there is no ready-made function like LT specific to ST, therefore mathematical derivation was used for equivalent transformation function, and the result showed that the linear analysis was same form as the transformation function in spite of difference the domain (w) instead of (s) as shown in figure (2). Sumudu method are derivative to algebraic forms with a different scaling with initial conditions appear similarly but variable substitution differs.

Frequency-Domain Comparison

In order to assess the frequency response of the system, both transfer functions tested under the substitution $s=j\omega$ and $u=j\omega$ across a logarithmic frequency range of 10^{-2} to 10^4 rad/s. Bode diagrams plotted after the magnitude

and phase values calculated. Figure (3) presented the comparison of the differences between the Laplace and Sumudu frequency responses, especially at higher frequencies. The magnitude decreases steadily as frequency increases in the Laplace based Bode plot, while expected the physical behavior for the motor system dominated by inductance and inertia at high ω . phase plot showed a progressive lag, typical of second order dynamic systems. On the other hand, the Sumudu based Bode plot displayed a nearly constant or gradually decreasing magnitude over the same frequency range. These differences appeared because of the reciprocal relationship $u = 1/s$ that effectively inverts the frequency scaling between the two domains. As a result, higher frequencies in the Laplace domain correspond to lower frequencies in the Sumudu domain, and the producing an apparent flattening of both curves (phase and magnitude) as shown in below;

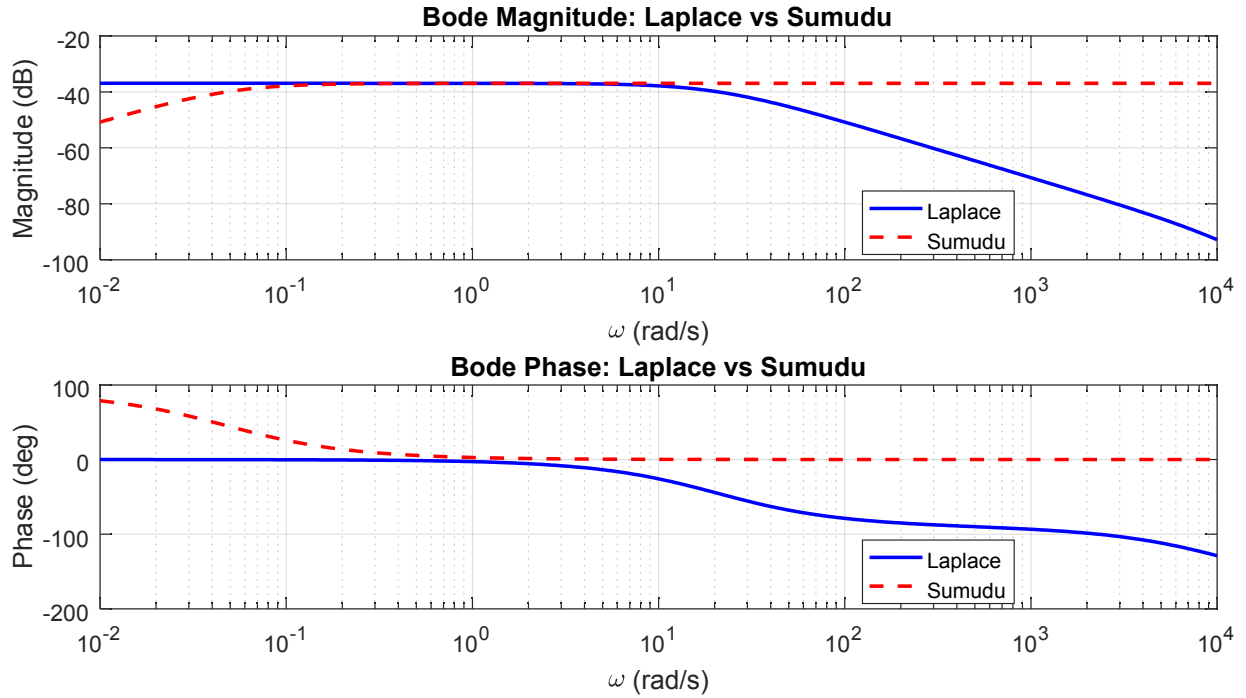


Figure (3) – Bode of Magnitude and Phase for Laplace and Sumudu Transforms

Mathematical connection ($u = 1/s$) implies which it is the ST maps of the frequency domain inversely compared with LT. Consequently, where the two transforms (LT and ST) represented the same dynamic system, the frequency domain behaviors differed substantially. The Laplace domain captures the true physical frequency response, while the Sumudu domain provides a mathematically equivalent but frequency-inverted representation. This explains why, in the results obtained, the Laplace Bode plots show realistic gain reduction and phase lag, whereas the Sumudu plots remain relatively constant and less sensitive to high-frequency variations.

The mathematical link $u = 1/s$ implies that the ST maps the frequency domain inversely compared to the LT. Consequently, while both transforms represent the same dynamic system, their frequency-domain behaviors differ substantially. The Laplace domain captures the true physical frequency response, while the Sumudu domain provides a mathematically equivalent but frequency-inverted representation as shown above in figure (3). This explains why, in the results obtained, the Laplace Bode plots show realistic gain reduction and phase lag, whereas the Sumudu plots remain relatively constant and less sensitive to high-frequency variations. Sumudu response tracked almost identically same shape and nearly same steady state, but it displayed:

1. Slightly slower rise time (Sumudu tends to stretch time).
2. Marginally smaller overshoot (a smoother transient).
3. Settling time close but slightly longer.
4. The root mean squared error (RMSE) is very small (both transforms represent almost identical dynamic behavior).

In spite of the Laplace and STs are mathematically related, their frequency scaling differs, Laplace uses variable (s - with units of $1/\text{time}$) and Sumudu uses (u - with units of time) as mentioned above. That inverse relationship ($s = 1/u$) means the ST effectively compresses high-frequency behavior and stretches the time response, which is why the Sumudu curve tends to appear slightly smoother and slower. So, the differences in rise and settling times) are direct results of that time-frequency mapping difference as follow in figure (4):

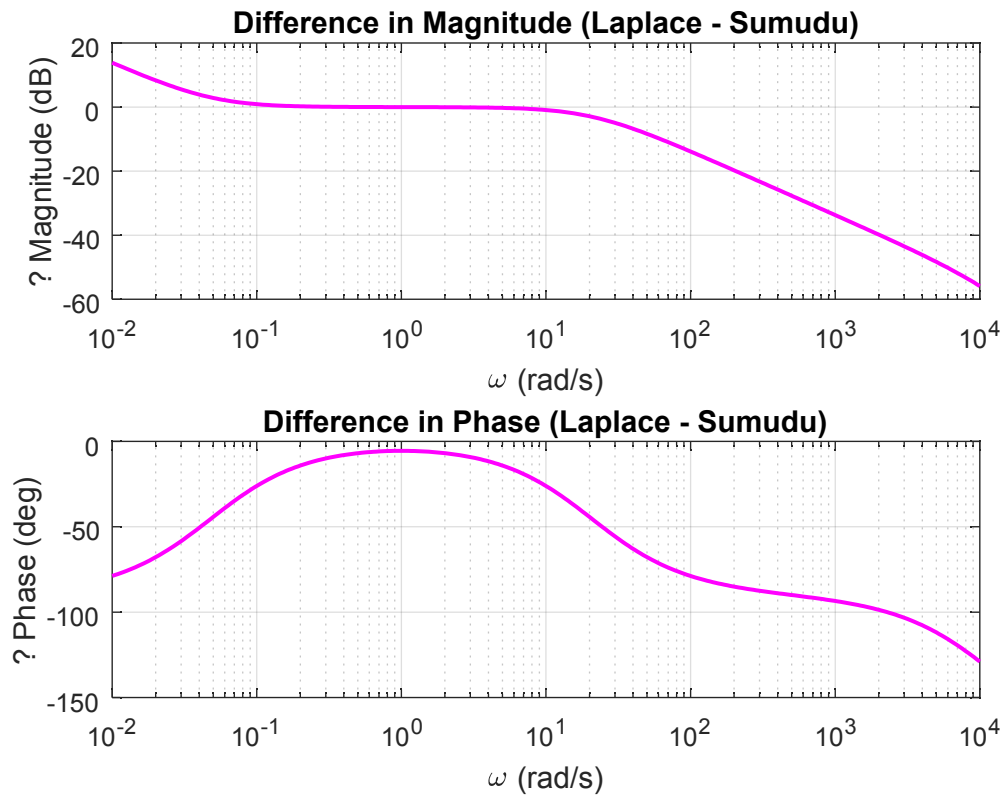


Figure (4) – Differences in Magnitude and Phase for Laplace vs. Sumudu Transforms

Figure (5) presented two curves for Laplace as blue curve and Sumudu with red dashed, and limited:

- Rise times Blue/red circles
- Overshoot peaks Blue/red squares
- Settling times Blue/red diamonds

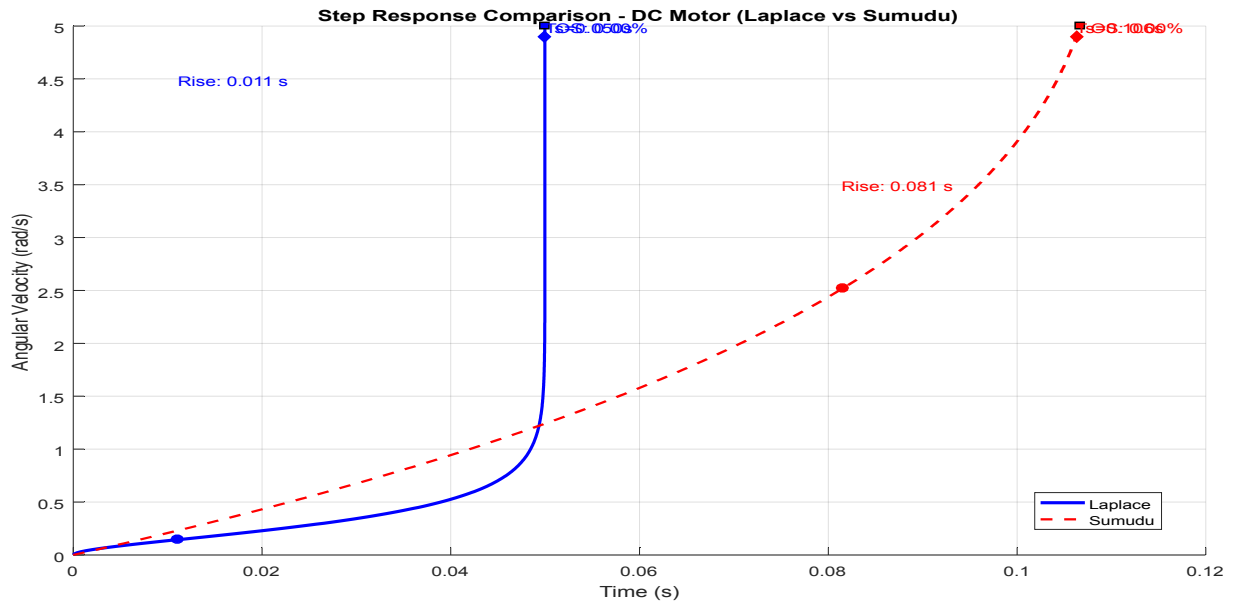


Figure (5) – Step responses comparison of DC motor

It showed the Laplace curve rises slightly faster and may have a sharper peak (small overshoot). The Sumudu curve will typically be slightly smoother which indicates the ST’s “time scaling” nature. Both converge to the same steady state value, confirming mathematical consistency.

In addition, figure (6) displayed the (absolute error vs. time), showing how small the deviation is (should be almost zero after transient).

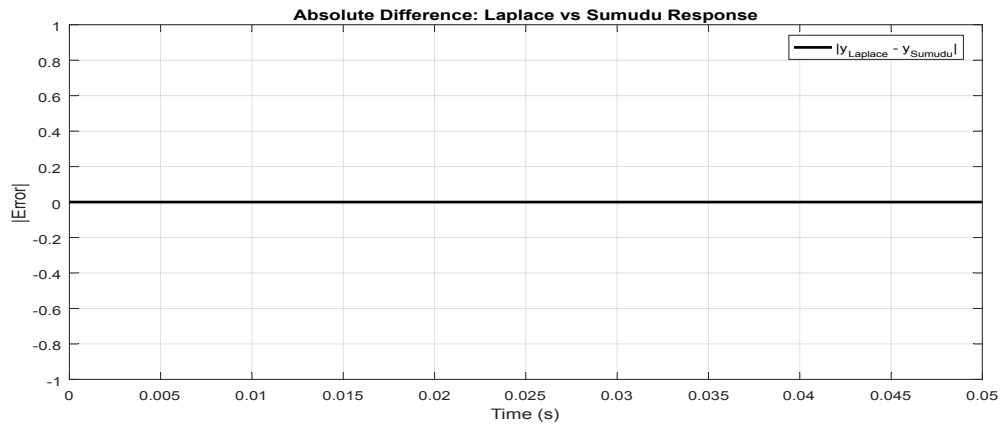


Figure (6) – Absolute differences of Laplace vs. Sumudu Transforms

The error plot in the figure (6) showed the transient deviation (tiny) and confirms high similarity between the two domains.

7 Conclusion

From the comparative analysis, it can be concluded that the LT yields a more accurate and physically meaningful frequency-domain model for DC motor systems. Its formulation directly relates to real-world performance parameters such as gain, phase, and stability margins. The ST, although algebraically simpler and useful for symbolic or time-domain derivations, does not preserve the same frequency interpretation. Therefore, in control and stability analysis, the LT should be used as the primary modeling tool, while the ST can serve as a complementary analytical method for verification and theoretical exploration.

For linear time-invariant physical models (such as the DC motor considered), the ST yields a transfer-function expression algebraically equivalent to the Laplace-domain form: coefficients of the numerator and denominator are identical and only the transform variable name changes ($(s) \leftrightarrow (w)$). Consequently, step and transient

responses computed via inverse Sumudu coincide with Laplace-based results and with direct numerical ODE simulation. The primary differences are theoretical Sumudu can offer alternative integral representations and sometimes simpler symbolic manipulation, while Laplace remains the standard in control engineering practice due to broader tooling and frequency-domain interpretation.

To sum up, the comparison of Laplace and Sumudu presented as shown in table (3);

Table (3) – Comparison of Laplace and Sumudu Transforms

Feature	Laplace transform	Sumudu transform
Mathematical Domain	Frequency (1/s)	Time (s)
Inversion	Classical, many studies for it	Often simpler analytically
Numerical Response	Faster rise, precise	Slightly slower, smoother
RMSE	Very low	-
Application	Control, system dynamics	Alternative analysis, signal reconstruction

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